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Research paper

Deep learning based bias correction model for numerical simulations of wave spectra

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ABSTRACT

To accurately characterize the energy distribution under different frequency ranges, this paper developed a deep learning-based wave spectral bias correction model for numerical simulations of wave spectra, which addresses the shape of the numerically predicted spectra. This model preprocesses the measured wave spectra using a Gaussian filter, and the frequency of the numerical simulation results from WAVEWATCH III is matched using the interpolation method. Afterwards, the simulated wave spectra serve as the input to the proposed model, and discrepancies between the numerical simulation results and buoy measurement data are fitted using the neural network model. Moreover, the correction performance of the model is validated using measured data in terms of spectral shape, significant wave height, peak frequency, and peak energy density. With the proposed method the mean absolute percentage error of those parameters of decreases by 5 %–15 % and the spectral correlation for the Hawaiian waters is no less than 0.93. The results demonstrate that the proposed model can more accurately characterize the wave energy distribution and significantly improve the accuracy of wave simulations.

1. Introduction

The wave spectrum is an essential tool for describing wave characteristics and is widely applied in fields such as ocean dynamics research and marine engineering design. It reflects the energy distribution of waves across different frequency and wavenumber spaces. Additionally, the wave spectrum can reveal the components of waves, including the generation and development of local wind waves, the propagation of distant swells, and the interactions between multiple wave systems. Furthermore, information such as the energy peak position and spectral width within the wave spectrum can further reflect the typical characteristics of waves, providing a basis for forecasting wave height, period, and other parameters. An accurate description of the wave spectrum provides more detailed results for describing the wave conditions (Lucas et al., 2011, 2023) than the simple information based on significant wave height and characteristic period. It can also support research on ocean climate change, ship navigation safety, and related studies.

Wave numerical models solve ocean waves using wave action balance equation (Whitham, 1965; Bretherton and Garrett, 1968). The development of third-generation wave models began with the WAM model (The WAMDI Group, 1988), which provided the first framework for resolving wave–wave interactions explicitly. They describe the input, propagation, and dissipation of wave energy. WAVEWATCH III (WW3) and SWAN, as typical representatives of third-generation wave models, are widely used for numerical simulation of ocean waves (Abbasi and Sheikh Bahai, 2025; Shankar et al., 2025).

In the wave model calculation process, the wind field forcing (Shankar et al., 2025; Ponce de Leon and Guedes Soares, 2008; Van Vledder and Akpınar, 2015), the source term methods (Beyramzadeh and Siadatmousavi, 2022; Zieger et al., 2015), and the grid discretisation (Campos et al., 2022; Oladejo et al., 2025) all affect the wave computation. Based on wave models, Ahn et al. (2022), Feng et al. (2020), and Ponce de León and Osborne (2020) have analysed the simulation accuracy of wave spectra.

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Stopa et al. (2016) evaluated wave spectrum models against satellite and buoy observations, analyzing the simulation accuracy under different physical parameterizations. They found that all models performed well in representing low-order wave moments, but their ability to reproduce directional spreading remained limited and requires improvement.

Amarouche et al. (2023) compared the wave spectrum simulation performance of the SWAN and WW3 models for the eastern Black Sea. They validated the simulation accuracy of both models across different frequency domains using annual and seasonal average scales. The results indicated that the WW3 model exhibited higher simulation accuracy in the frequency range of 0.13 Hz $< f <$ 0.17 Hz. However, issues such as the underestimation of low-frequency energy and the overestimation of high-frequency energy remained. Additional results concentrating on the more difficult extreme conditions can be found in Bernardino et al. (2021).

Elshinnawy et al. (2024a, 2023) conducted long-term wave hindcast studies exceeding 50 years for the Red Sea and the Mediterranean using the WW3 model with ERA5 and UERRA-MESCAN-SURFEX wind fields, respectively, as input. They analysed the seasonal and interannual variations of waves, long-term trends of wind-driven waves and swells, and specifically focused on wave behavior under extreme weather conditions. Elshinnawy et al. (2024b) evaluated the wave simulation performance using the WW3 model with ERA5 wind fields as input. In addition, a parameterisation was developed for WW3 to correct biases in ERA5 wind speeds, enabling high-accuracy modeling of waves under extreme conditions.

Wave models describe the wave propagation process using different source functions during the solution of wave dynamics. The simulation accuracy is often influenced by the simplification of physical processes, source term parameterisation, and uncertainties in boundary conditions, which leads to differences between the simulated results and the actual marine environment (Zhang et al., 2024).

As a crucial means of obtaining measured wave information, buoy measurements are widely used for validating numerical model results due to their high observational accuracy (Sun et al., 2019; Bennett and Mulligan, 2017). They are also extensively applied in studies ranging from ocean climate analysis (Jamous and Marsooli, 2023; Gemmrich et al., 2011) to local area studies (Rusu and Guedes Soares, 2014). Buoys, equipped with accelerometers and gyroscopes, collect time series data of vertical displacements and horizontal surface movements to capture the dynamic characteristics of wave fluctuations. Through data quality control, methods such as correlation functions, maximum entropy, and fast Fourier transform are further utilised to compute the wave spectrum. Additionally, buoys can extract key parameters such as significant wave height, period, and propagation direction by recording the wave variation process. The existing distribution of ocean buoys is sparse, with a concentration along continental coastlines.

Currently, publicly available buoy data are provided by several major centres, including the National Data Buoy Center (NDBC), the Copernicus Marine Environment Monitoring Service (CMEMS), and the Japan Meteorological Agency (JMA). Among them, NDBC operates under the National Oceanic and Atmospheric Administration (NOAA) of the United States. The time resolution of the wave data can reach up to 0.5 h, with a measurement accuracy of 0.2 m for wave height.

However, buoys face challenges such as high cost and deployment difficulties. As a result, their distribution is sparse, and their spatial coverage is relatively limited. This makes continuous global ocean observation difficult. Furthermore, buoy data are often subject to loss or interruptions due to factors such as equipment ageing, communication failures, and extreme weather events. This affects both data continuity and overall reliability.

Environmental information computed by wave models has been improved using buoy observational data. The results better reflect the actual oceanic and meteorological conditions by incorporating discrete observational data into numerical models. Common data assimilation methods, such as variational assimilation, Kalman filtering, and optimal interpolation, have been applied (Almeida et al., 2016; Zou, 2025). Jiang et al. (2025) optimised the physical parameterisation schemes of WRF and SWAN simulations for Typhoon cases in the South China Sea using assimilation techniques. The results showed that, compared to the default scheme, this method effectively improved the wave simulation outcomes. After assimilation, the RMSE of the significant wave height was 0.322 m. For typhoon waves with return periods of 100 and 50 years, the improvements in wave height were 1.12 m and 0.96 m, respectively. Wang et al. (2024) corrected the WW3-simulated wave results using wave spectrum data observed by the Surface Wave Investigation and Monitoring instrument aboard the China-France Oceanic Satellite (CFOSAT) satellite. Hasselmann et al. (1997) proposed a wave spectrum assimilation method based on optimal interpolation. The wave spectrum was decomposed into different wave systems and represented by three characteristic parameters: significant wave height, average propagation direction, and average frequency. Radar data were used to optimise these parameters, enabling wave spectrum assimilation.

In recent years, with the continuous advancement of artificial intelligence technology, machine learning methods have been increasingly applied in marine environments. Deep learning methods have been applied in recent years to optimise existing wave data.

Deshmukh et al. (2016) developed an artificial neural network (ANN)-based data assimilation method to enhance the forecast accuracy of significant wave height and wave period in numerical wave models. Zhang et al. (2024) developed the WW3-ANN model using deep learning methods. The model corrects the significant wave height by addressing the differences between buoy observations and simulation results.

da Silva et al. (2025) optimised the wave height of the WW3 model based on the Long Short-Term Memory (LSTM) model. The model was trained using wave height, wind speed, and coordinate location to achieve high-precision wave forecasting for the Brazilian coastal region. Costa et al. (2023) also used the LSTM model and compared its performance with that of the multilayer perceptron.

Henriques et al. (2025) employed five different AI methods to enhance wave forecasts using buoy data, offering a comprehensive comparison of the methods' capabilities with LSTM. This approach solves the problem of balancing accuracy and efficiency in traditional models.

Lucero et al. (2023) improved the post-forecast results of the WW3 model using a machine-learning model. By using the wave spectrum as input, they reduced the calculation errors for significant wave height, mean period, and mean wave direction.

Hwang et al. (2024) proposed a wave spectrum computation model based on a Deconvolutional Neural Network, in which wave parameters such as wave height and wave period were used as model inputs. The model results were validated against buoy-observed spectra, and the accuracy was found to be comparable to that of the JONSWAP spectrum. Zhang et al. (2024) developed a wave spectrum correction method using a Vision Transformer (ViT) deep learning model to adjust the CFOSAT satellite-observed spectra. This approach resolved issues in the original SWIM spectra, including speckle noise, low-frequency interference peaks, and missing short-wave data.

However, traditional data assimilation methods rely on ocean observation information. When observational data is unavailable, especially in remote areas or under extreme weather conditions, the applicability of conventional assimilation methods is challenged. Therefore, optimising and correcting ocean environmental data in the lack of sufficient observational data has become an important research direction for improving the accuracy of ocean environment forecasting (Emmanouil et al., 2010). The current deep learning-based wave correction methods primarily focus on global wave parameters, such as significant wave height and wave period, whereas corrections to the wave spectrum rely mainly on spectral parameters.

A wave spectral bias correction model (E_f-*ANN*) is constructed in this paper based on the advantages of deep learning methods in large-scale data processing, complex feature extraction, and non-linear modelling. The model utilises buoy measurements to identify discrepancies between the simulated and real data, thereby achieving an effective correction of the wave spectrum. This method addresses the error issues caused by the simplification of physical processes in traditional methods. It also overcomes the limitations of data assimilation methods that are constrained by the availability of measurements. Furthermore, the model's correction performance is evaluated through multiple metrics, including the wave spectrum and key spectral parameters such as significant wave height, peak frequency, and peak value. The model's applicability is also explored by utilising multiple buoy locations within the sea area.

The organisation of this paper is as follows: Section 2 introduces the data used in this study. Section 3 presents the wave numerical simulation method based on the WW3 model. Section 4 describes the wave spectrum bias correction method based on deep learning. Section 5 analyses the correction performance and adaptability of the model. Section 6 concludes the work presented in this paper.

2. Datasets

The data utilised in this study include wind field, topography, and buoy measurement data.

The European Centre for Medium-Range Weather Forecasts (ECMWF) developed the ECMWF Reanalysis v5 (ERA5) dataset, which is utilised in this study to provide wind field information. ERA5 combines model data with observations from around the world to create a globally complete and consistent dataset, using the laws of physics. The dataset encompasses atmospheric, oceanic, and land variables dating back to 1940. For the wind field, ERA5 provides an hourly temporal resolution and a spatial resolution of 0.25°.

The topographic data are obtained from the ETOPO1 dataset developed by the National Geophysical Data Center (NGDC). This dataset integrates multi-source observations, including satellite altimetry, sonar sounding, and land survey data, to generate high-resolution global topographic information.

The buoy measurement data is sourced from the National Data Buoy Center (NDBC). NDBC is part of the National Oceanic and Atmospheric Administration (NOAA). It provides and maintains observational data from over 100 buoys and coastal stations worldwide. The provided data includes ocean wave parameters (wave spectrum, significant wave height, and wave period) and meteorological variables (wind speed, wind direction, air temperature, and atmospheric pressure). The wave spectrum data is updated every 0.5/1 h, ranging from 0.02 Hz to 0.485 Hz.

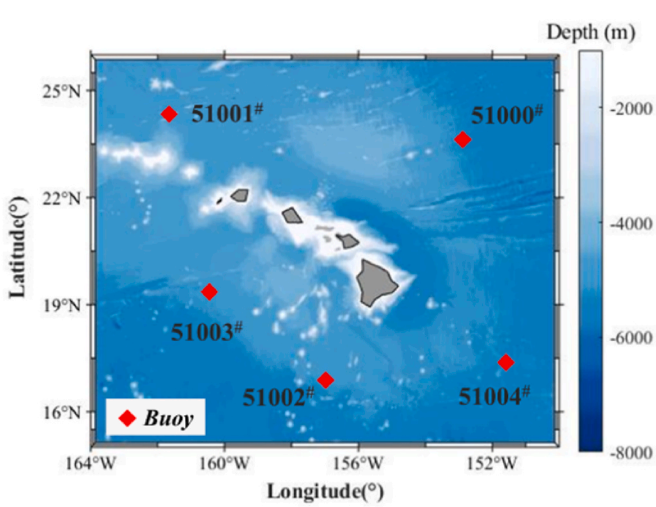

Fig. 1. Schematic diagram of the buoy locations.

Five buoys (51000#-51004#) are selected for the Hawaii region, with specific information shown in Fig. 1 and Table 1. The year's wave data from 2022 are used to test and validate the model. Due to equipment issues and extreme weather conditions, some periods have missing data. Over 8000 data sets are available for the above buoy locations throughout 2022.

3. Wave numerical simulation based on the WW3 model

3.1. Principles of the WW3 model

For the numerical simulation of large-scale waves, wave models based on the spectral energy equation can accurately describe wave generation, propagation, and dissipation. As a representative third-generation wave model, WW3 has been widely used in global and regional wave simulations, marine hazard warnings, and climate research due to its computational efficiency.

Wave energy (E) represents the total energy of the waves per unit area. By normalising wave energy with respect to frequency, wave action (A) is obtained. In non-quiescent water environments, the work done by the current is on the mean momentum transfer of waves. Under the influence of the current, frequency changes lead to the non-conservation of E, while A remains conserved. In the WW3 wave model, A and E are described in terms of their distributions in wave number-direction space through the wave action density spectrum (N) and the wave energy density spectrum (F), respectively.

$$A \equiv \frac{E}{\sigma} \tag{1}$$

$$N(k,\theta) \equiv \frac{F(k,\theta)}{\sigma} \tag{2}$$

In Eqs. (1) and (2), $\sigma = 2\pi f_r$ represents the frequency, k denotes the wave number, and θ represents the wave direction.

The WW3 model numerically solves wave dynamics based on the wave action density spectrum. The calculation process incorporates the effects of wind fields, ocean currents, sea ice, and seabed topography. The physical processes are represented through source term parameterisation (Whitham, 1965; Bretherton and Garrett, 1968).

$$\frac{DN}{Dt} = \frac{S}{\sigma} \tag{3}$$

$$S = S_{ln} + S_{in} + S_{nl} + S_{ds} + S_{bot} + S_{ice} \cdots \tag{4}$$

In Eqs. (3) and (4), S represents the source terms, where S_{ln} is the linear input term, S_{in} is the wind energy input, S_{nl} denotes non-linear wave-wave interactions, S_{ds} represents the dissipation term, S_{bot} is the wave-bottom interaction term, and S_{ice} represents wave-ice interaction term.

3.2. The WW3 model setup and calibration

This subsection employs the WW3 wave model to conduct a one-year numerical simulation of waves in the Hawaiian waters. Located in the central Pacific Ocean, this area is characterised by wide-open waters and complex sea states influenced by both local wind fields and remotely

Table 1
Buoy locations.

No.	Buoy Station	Latitude (°)	Longitude (°)
1	51000	23.528	153.792
2	51001	24.453	162.000
3	51002	17.043	157.742
4	51003	19.196	160.639
5	51004	17.533	152.255

generated swells. During the WW3 simulations, the ERA5 dataset provides wind forcing, while the ETOPO1 dataset supplies bathymetric information.

Distant waves affecting the Hawaiian waters (15°N–26°N, 164°W–150°W) are considered using a nested approach. The size of the boundary sea area is important. A small boundary cannot account for long-distance swell propagation, while a large boundary increases computational cost and simulation time. Three schemes are designed to investigate the impact of boundary size on wave simulation accuracy in the target sea area, as detailed in Table 2.

The influence of the three boundary schemes on wave simulation results is investigated through a boundary-domain size convergence analysis, as shown in Fig. 2.

The comparison shows that *WW3-B*3 better represents the wave conditions within the sea area, with simulated wave results in closer agreement with buoy measurements. Therefore, enlarging the boundary domain can improve the accuracy of wave simulations. However, a larger boundary domain is not always better, as it requires higher computational resources. The results show that the current scheme *WW3-B*3 adequately represents the wave conditions in the study area. Therefore, this scheme is adopted to configure the boundary conditions for WW3 simulations in the Hawaiian region. Fig. 3 illustrates the specific nested configuration used in the WW3 simulations.

The WW3 model represents wind energy input, energy dissipation, and wave–wave energy transfer through source terms. To evaluate the effects of wind input and dissipation on wave simulations in the Hawaiian region, the ST3 and ST4 schemes proposed by Ardhuin et al. are applied. The BJA scheme, which performs better under ST3, and the T471 scheme, which performs better with ECMWF wind forcing under ST4, are selected. The detailed parameter settings are provided in Table 3. In addition, the Discrete Interaction Approximation (DIA) proposed by Hasselmann et al. (1985) is applied to simulate nonlinear wave–wave interactions, with λ_{nl} set to 0.25 and C set to 2.5×10^7. Linear energy input is represented using the LN1 scheme of Tolman et al. (1992) to compute wave growth under initial conditions. the bed friction induced wave energy dissipation is modeled using the SHOWEX formulation of Ardhuin et al. (2003). Moreover, as the Hawaiian region lies at the boundary between tropical and subtropical zones, sea ice effects on waves are not considered.

The results of two source term configurations are compared using buoy-observed wave heights, as shown in Fig. 4. It can be seen that the *WW3-S*2 scheme aligns more closely with the buoy observations. Therefore, the ST4 scheme is selected for wave simulations in the Hawaiian region.

The domain of the wave spectra is discretised into 36 directions, with 10° bins, and 47 frequencies spaced nonlinearly from 0.04118 Hz to 0.9254 Hz, employing a frequency increment factor of 1.07. The model requires the definition of four time steps: the maximum global time step ($dt_{\max}$), the maximum CFL time step for x-y (dt_{xy}), the maximum CFL time step for k-th ($dt_{k\theta}$), and the minimum source term time step ($dt_{\min}$). These are set to 1800s, 600s, 900s, and 30s, respectively. This configuration defines the spatiotemporal discretisation settings for the WW3 simulation in the Hawaiian region for the period from January 1 to December 31, 2022.

Table 2
Boundary sea area configuration schemes.

No.	Boundary Sea Area	Target Sea Area
*B*1	–	15°N-26°N; 164°W-150°W
*B*2	0°N-41°N; 180°E–135°W	
*B*3	20°S-61°N; 160°E–115°W	

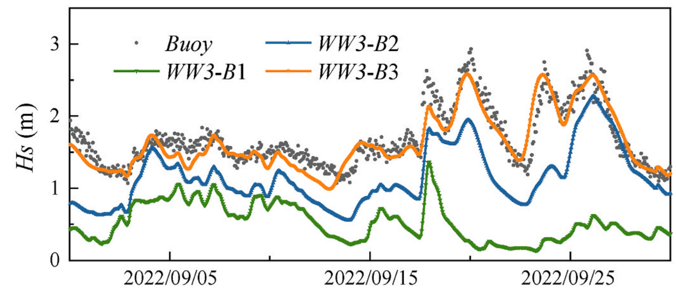

Fig. 2. WW3 simulation results under different boundary sea area at *Buoy* 51000.

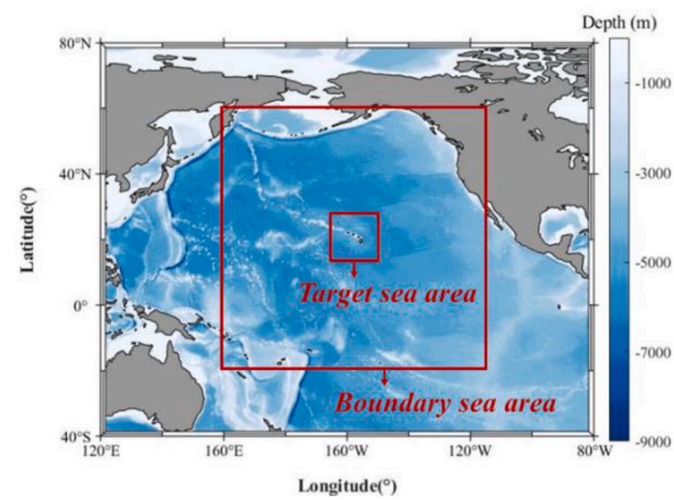

Fig. 3. Boundary condition setup scheme.

Table 3
Source term configuration schemes.

No.	S_{in} + S_{ds}	Parameter settings
*S*1	ST3	BJA: $z_u = 10$; $\alpha_0 = 0.0095$; $\beta_{\max} = 1.2$; $p_{\text{in}} = 2$; $z_\alpha = 0.011$; $s_1 = 0$
*S*2	ST4	T471: $z_u = 10$; $\alpha_0 = 0.0095$; $\beta_{\max} = 1.43$; $p_{\text{in}} = 2$; $z_\alpha = 0.006$; $s_1 =$ 0.66

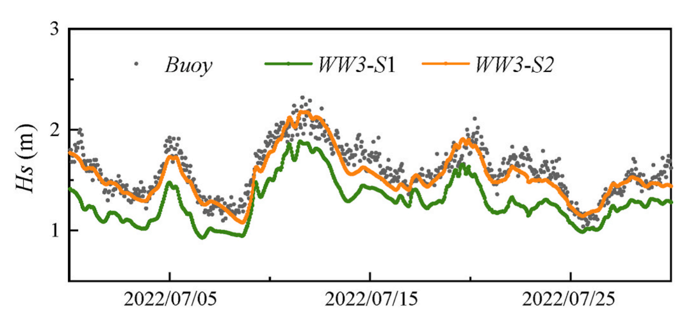

Fig. 4. WW3 simulation results under different source term schemes at *Buoy* 51000.

4. Wave spectrum bias correction method based on deep learning

4.1. The E_f -ANN model

Artificial Neural Network (ANN), proposed by McCulloch and Pitts in 1943, is a data model inspired by the human brain's neural network. It is widely used in tasks such as pattern recognition, data classification, regression prediction, and optimisation. The model transmits data through neurons and consists of input, hidden, and output layers. ANN learns the non-linear relationships between data by continuously optimising the connection weights between neurons in each layer.

The structure of the ANN is shown in Fig. 5, and it can be represented by

$$y_j^l = F\left(\sum_{k=1}^{m} w_{jk}^l x_k^{l-1} + b_j^l\right) \tag{5}$$

In Eq. (5), x represents the model input, y represents the model output, l represents the current layer, j represents the neurons in the current layer, k represents the neurons in the upper layer, m represents the number of neurons in the upper layer, w represents the weights corresponding to each input element in the current neuron, and b represents the bias of the neuron.

This study develops the E_f-*ANN* model based on the ANN to correct the wave energy spectrum (hereinafter denoted as E_f). The model targets the wave spectrum calculated by the wave model, using buoy measurements as a reference. The model effectively corrects the simulated wave spectrum by modelling the discrepancies between the two. This addresses the issue of differences between simulation results and actual wave conditions caused by the simplifications in physical process descriptions when solving wave dynamics numerically in models like WW3. The specific structure of the model is shown in Fig. 6.

The E_f-*ANN* model constructed in this study consists of three components: Wave Model, Buoy Measurement, and ANN Correction.

The Wave Model employs numerical wave models to generate a simulated wave spectrum dataset (E_{f_WW3}). This module utilises wave models to numerically solve the wave energy spectrum within the target domain based on ocean surface wind fields and bathymetric information. To address the discrepancies between numerically simulated and buoy-measured wave spectra in the frequency domain, linear interpolation is applied to unify their spectral resolutions. The calculation method is as follows:

$$E(f_{ndbc}) = E(f_i) + \frac{E(f_{i+1}) - E(f_i)}{f_{i+1} - f_i} \cdot (f_{ndbc} - f_i) \tag{6}$$

In Eq. (6), f_{ndbc} represents the buoy observation frequency, f_i denotes the WW3 simulated spectrum frequency, and E stands for the wave spectrum. After frequency alignment, the wave spectrum frequency range is 0.0425–0.485 Hz, with a total of 44 frequency bins. In addition, considering the issues of missing or incomplete buoy data, the time series of wave spectra at specific locations is preprocessed using observational data as a reference. Finally, E_{f_WW3} serves as the input for the ANN Correction module.

The Buoy Measurement module constructs an observed wave spectrum dataset (E_{f_Buoy}) based on buoy data. Observational records from NDBC and other sources are processed using Gaussian filtering (Ito and Xiong, 2000). This smooths out abnormal fluctuations in the raw data, resulting in a preliminary smoothed spectrum. A peak restoration method is applied to preserve the original spectral shape's dominant

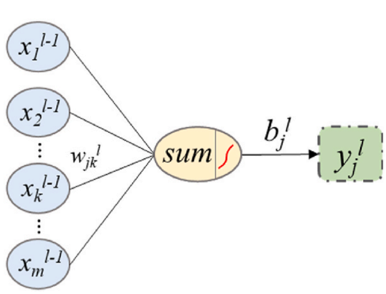

Fig. 5. Schematic diagram of interneuronal information transmission.

frequency and energy distribution characteristics. The specific calculation method is as follows:

$$G(t) = \frac{1}{\sqrt{2\pi s^2}} \exp\left(-\frac{t^2}{2s^2}\right) \tag{7}$$

$$y_G(n) = \sum_{m=-k}^{k} y_0(n+m) \cdot G(m) \tag{8}$$

$$y_{0\_\max} = y_0(n_p) \tag{9}$$

$$y(n) = \begin{cases} \beta \cdot y_0(n_p) + (1-\beta) \cdot y_G(n_p), n = n_p \\ y_G, n \neq n_p \end{cases} \tag{10}$$

In Eqs. (7)–(10), $G(t)$ represents the Gaussian kernel. $G(m)$ denotes the discretised and truncated Gaussian kernel, typically represented as a symmetric sequence of length $2k+1$. y_0 denotes the original signal, and y_{0_max} refers to the peak value in the original signal. y_G represents the output signal after Gaussian filtering, while y is the final output after peak recovery. t is the independent variable, n indicates the index position, m represents the sliding window offset, s denotes the standard deviation of the Gaussian kernel, k is the Gaussian kernel radius, and β is the weighting coefficient for peak recovery.

For the buoy-observed wave spectrum, a Gaussian kernel $G(x)$ is constructed using Eq. (7), generating a symmetric, centrally focused weighting function, as shown in Fig. 7. Eq. (8) applies convolution smoothing to the original signal. A symmetric window with radius k is used for local sliding weighted averaging of the observed wave spectrum y_0, performed through the Gaussian kernel, which removes abnormal fluctuations and observational errors. The filtered signal y_G is obtained, smoothing the observed wave spectrum. Eq. (9) extracts the peak values y_{0_max} from the observed signal, retaining the dominant wave energy information. During the filtering process, extreme values are also smoothed, reducing the energy peak. Eq. (9) identifies and records the original spectrum's peak positions and corresponding energy levels. Eq. (10) restores the wave spectrum peaks. A weighting factor is introduced to merge the original values $y_0(n_p)$ and the filtered values $y_G(n_p)$ at the peak points, balancing the retention of the energy peak and the overall smoothing characteristics. The final output, $y(n)$ is a smoothed signal with high fidelity.

Additionally, data gaps resulting from extreme weather events or equipment failures are removed from the dataset. Finally, E_{f_Buoy} serves as the output for the ANN Correction module.

The ANN Correction module corrects the wave spectrum (E_f). The module takes the E_{f_WW3} as input and outputs the E_{f_Buoy}. Both E_{f_WW3} and E_{f_Buoy} have a data dimension of 44×6324, where 44 is the number of frequency bins, and 6324 represents the number of time steps. The module consists of an input layer, hidden layers, an output layer, and the model output. The input and output layers contain 44 neurons, representing the wave spectrum results corresponding to 44 frequencies at a single time step. The model uses three fully connected hidden layers, with 200, 200, and 100 neurons, respectively. All hidden layers use the Rectified Linear Unit (ReLU) activation function to enhance the model's ability to express nonlinearity. The output layer employs a linear activation function.

The model processes the E_{f_WW3} in batches during parameter training, which is read and fed into the network one time step at a time for forward propagation, and outputs the corrected wave spectrum (E_{f_ANN}). The model employs the Mean Squared Error (MSE) as the loss function to measure the difference between the model output and the observed spectrum (E_{f_Buoy}). During optimisation, the Adam optimiser (Kingma, 2014) is employed with a learning rate of 0.001. The number of training iterations is set to 200. This ultimately enables the correction of the wave spectrum (E_f).

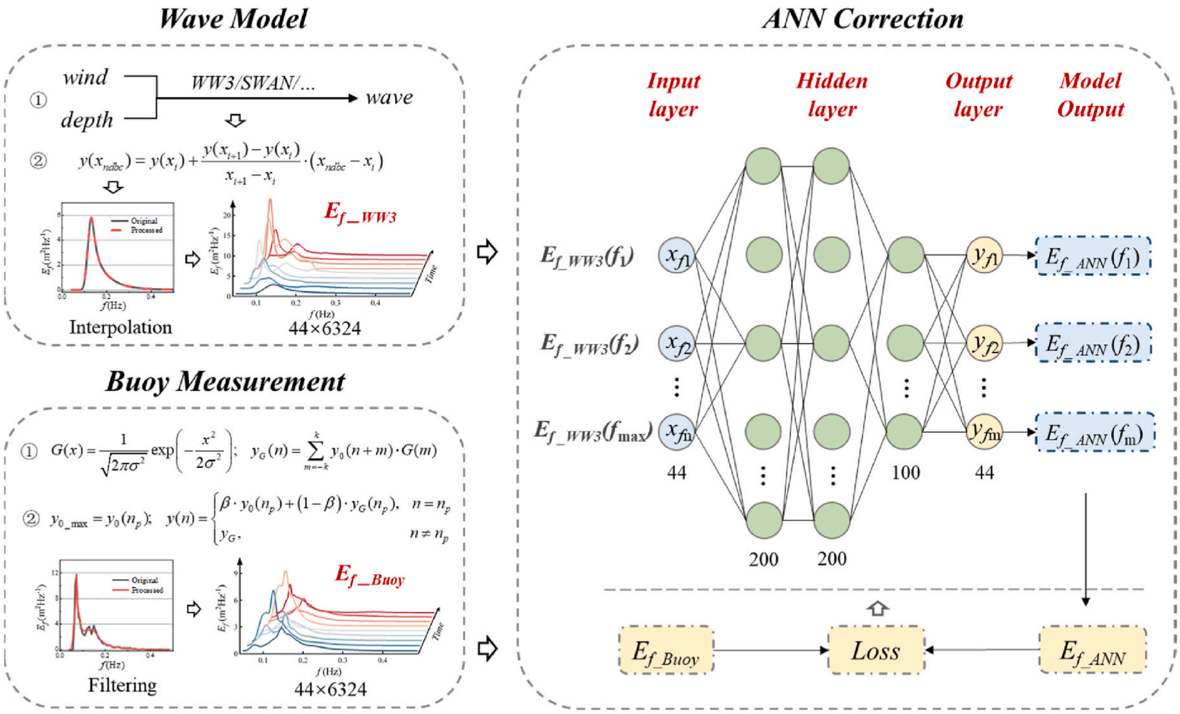

Fig. 6. Schematic diagram of the E_f-*ANN* model structure.

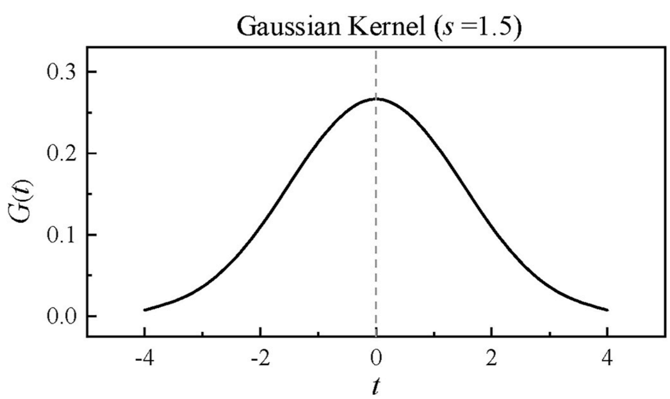

Fig. 7. Shape of the Gaussian kernel.

$$Loss = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{11}$$

In Eq. (11), n represents the number of samples, y_i and $\hat{y}_i$ represent the actual values(E_{f_Buoy}) and model predicted values(E_{f_ANN}), respectively.

In this study, wave spectra at five buoy locations provided by NDBC are updated every 0.5/1h. After temporal unification and missing-value processing, more than 8000 sets of simulated and observed wave spectra are available at each buoy location. For each buoy, 70 % of the spectra are randomly selected as training data, and the remaining are used as testing data to construct the dataset. The wave spectra at buoy 51000 are employed for model accuracy validation, while the other four buoys are used for model adaptability analysis. Ultimately, the E_f-*ANN* model is developed with E_{f_WW3} as input and E_{f_Buoy} as output. By constructing the bias between numerically simulated wave spectra and buoy measurements, the model enables the correction of wave spectrum simulations within the study area.

4.2. Evaluation metrics

To validate the model's accuracy, the mean absolute error (MAE), mean absolute percentage error (MAPE), root mean square error (RMSE), and Pearson's correlation coefficient (r) are used as the accuracy evaluation metrics in this study. The specific calculation methods are shown in Eq. (12)–(15).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - y'|\tag{12}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - y_i'}{y_i}\right|\tag{13}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - y_i)^2}\tag{14}$$

$$r = \frac{\sum_{i=1}^{n}\left(y_i - \frac{1}{n}\sum_{i=1}^{n}y_i\right)\left(y_i' - \frac{1}{n}\sum_{i=1}^{n}y_i'\right)}{\sqrt{\sum_{i=1}^{n}\left(y_i - \frac{1}{n}\sum_{i=1}^{n}y_i\right)^2\sum_{i=1}^{n}\left(y_i' - \frac{1}{n}\sum_{i=1}^{n}y_i'\right)^2}}\tag{15}$$

In the above equations, n represents the sample size, while y_i and y_i' represent the wave observation results and the wave calculation results, respectively.

In addition, in order to further assess the accuracy of wave spectrum calculation, the spectral shape parameters E_{max}, F_p, H_s are introduced to evaluate the spectral peak value and spectral distance.

$$E_{\max} = \max(E_f)\tag{16}$$

$$F_p = f_{E\max}\tag{17}$$

$$H_s = 4\sqrt{\int_0^{\infty} E_f df} \tag{18}$$

In Eq. (16)–(18), E_f represents the wave spectrum, H_s is the significant wave height, F_p is the peak frequency, and E_{max} is the wave spectrum peak value.

5. Results and discussion

5.1. Model accuracy

In this section, the accuracy of the E_f -*ANN* model is analysed. Four random time instances are selected to compare the wave spectrum results before and after correction by the model. The comparison includes the buoy-observed wave spectrum, WW3 numerical simulation wave spectrum, and the wave spectrum corrected by the E_f-*ANN* model at different times. The results are shown in Fig. 8.

The comparison shows that after correction by the E_f -*ANN* model, the wave spectra are closer to the buoy measurements. The WW3 numerical simulation wave spectra reflect the frequency distribution and overall trend of the wave spectrum well, but there are still some discrepancies between the simulated and observed peak values at certain times. The E_f -*ANN* model can effectively correct the wave spectra and reproduce the distribution of wave energy better.

The MAE, RMSE, MAPE, and r are shown in Fig. 9. At *Buoy* 51000#, the MAE, RMSE of the WW3 simulation are 0.62 m^2Hz^{-1}, 1.13 m^2Hz^{-1}, and the MAPE is 37.0 %. After correction by the E_f -*ANN* model, the MAE, RMSE and MAPE significantly decrease compared to WW3, with MAPE decreasing to 27.6 %. As shown in Fig. 10, the regression equation for WW3 compared to the observations is $y = 0.660x+0.145$, indicating a general underestimation in the computations. In contrast, the regression equation for E_f -*ANN* is $y = 0.856x+0.117$. The spectral correlation improves from 0.83 to 0.94. These results indicate that the E_f -*ANN* model effectively improves the simulation accuracy of the wave spectrum.

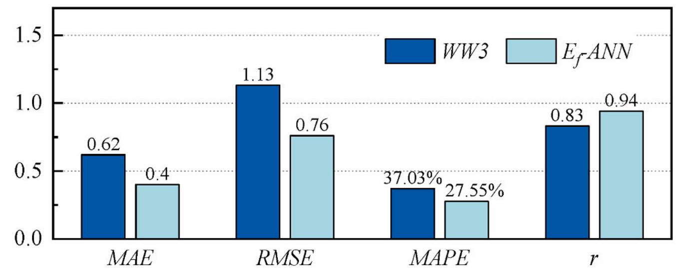

Fig. 9. Comparison of accuracy before and after the E_f -*ANN* model at *Buoy* 51000.

Moreover, the accuracy of waves in extreme sea conditions is more important. The correction effect of the E_f -*ANN* model in medium to high sea conditions was analysed, and the results are shown in Table 4.

By comparison, the E_f -*ANN* model shows higher accuracy in wave spectrum correction. The MAPE values under sea states of levels 6, 7, 8, and 9 are 25.1 %, 22.0 %, 22.8 %, and 18.8 %, respectively, indicating a significant improvement over the WW3 results. Additionally, the correction effect of the wave spectrum under sea states of levels 8–9 is demonstrated, as shown in Fig. 11. It can further be seen that the model achieves high accuracy in extreme sea states, reducing the underestimation of wave calculations and better capturing peak values during extreme storms.

To comprehensively evaluate the E_f -*ANN* model's correction performance, three parameters (i.e., H_s, F_p, and E_{max}) are used to calculate the accuracy of both the spectral width and the spectral peak. As shown in Table 5 and Fig. 12, the MAPE for H_s, F_p, and E_{max} calculated by the E_f -*ANN* model decreases from 16.4 %, 16.2 % and 28.2 % in WW3 to 11.1 %, 9.8 % and 24.1 %. In Figs. 12 and 10MAPE refers to 10 times the MAPE. The correlation coefficients (r) improve from 0.94, 0.645 and 0.88 to 0.97, 0.789 and 0.93, respectively. These results demonstrate

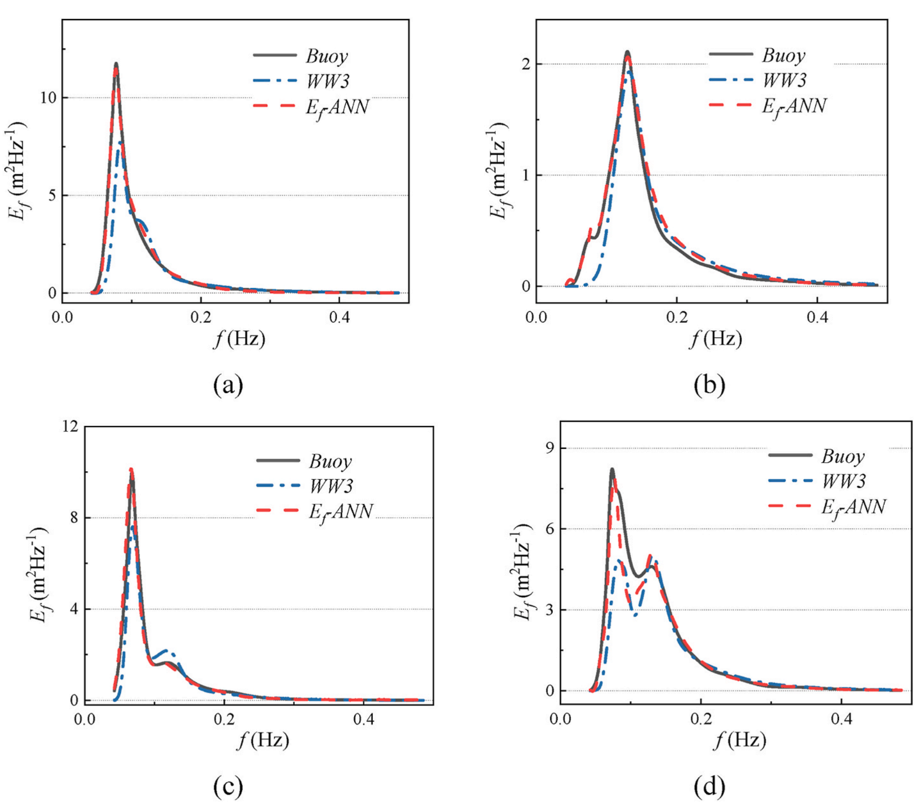

Fig. 8. Comparison of wave spectra before and after correction by the E_f -*ANN* model at *Buoy* 51000.

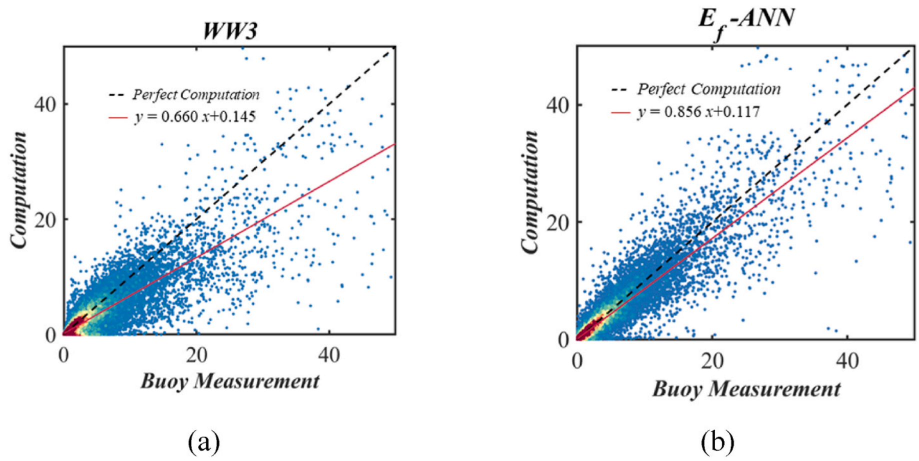

Fig. 10. Comparison of wave spectrum correlation before and after the E_f -*ANN* model at *Buoy* 51000.

Table 4
Comparison of the E_f correction effect under high sea ststes at *Buoy* 51000.

Sea State		WW3				E_f -ANN			
Level	*Hs*	MAE	RMSE	MAPE	r	MAE	RMSE	MAPE	r
6	2.0–3.0	0.52	0.98	33.5 %	0.87	0.38	0.74	25.1 %	0.94
7	3.0–4.0	1.62	3.00	33.8 %	0.85	0.93	1.84	22.0 %	0.95
8	4.0–5.5	3.58	6.30	33.5 %	0.91	2.20	4.24	22.8 %	0.96
9	5.5–7.0	5.12	8.90	31.8 %	0.91	2.89	5.40	18.8 %	0.95

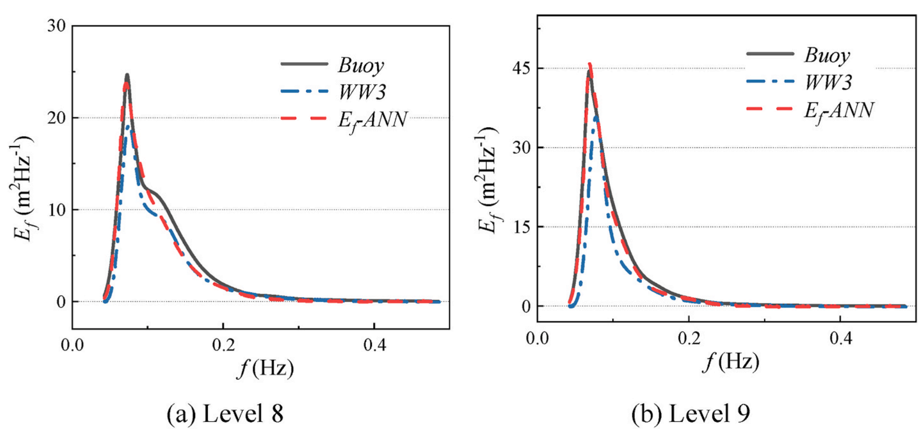

Fig. 11. Comparison of wave spectra before and after correction under medium to high sea states at *Buoy* 51000.

Table 5
Multi-metric combined validation at *Buoy* 51000.

	WW3				E_f -ANN			
	MAE	RMSE	MAPE	r	MAE	RMSE	MAPE	r
H_s	0.07	0.13	16.4 %	0.94	0.04	0.07	11.1 %	0.97
F_p	0.013	0.022	16.2 %	0.645	0.009	0.017	9.8 %	0.789
E_{max}	2.76	5.67	28.2 %	0.88	2.02	3.89	24.1 %	0.93

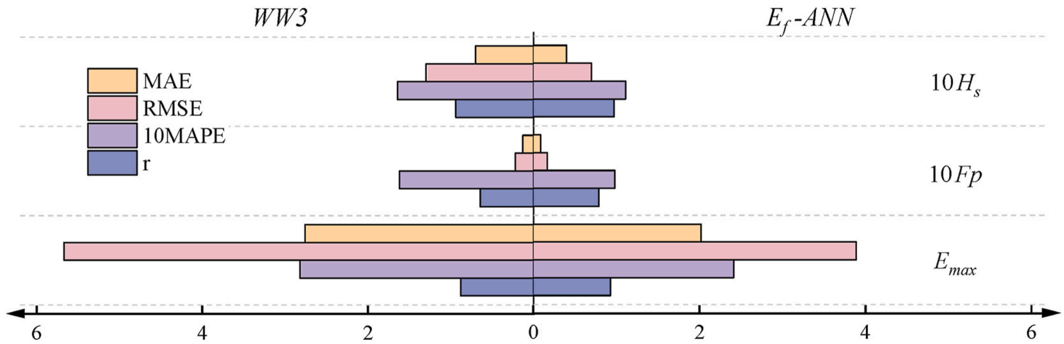

Fig. 12. Comparison of spectral shape parameter accuracy at *Buoy* 51000.

Table 6
Percentage decrease in MAPE (δ) at *Buoy* 51000.

Parameter	δ
E_f	25.6 %
H_s	32.3 %
F_p	39.3 %
E_{max}	14.4 %

that the E_f -*ANN* model effectively enhances the accuracy of spectral shape parameter calculations and enables comprehensive correction of wave information.

Next, the improvement in the calculation accuracy of the wave spectrum and spectral shape parameters has been analysed.

$$\delta = \frac{\left|MAPE_{WW3} - MAPE_{E_f-ANN}\right|}{MAPE_{WW3}} \tag{19}$$

In Eq. (19), δ represents the reduction in MAPE for each parameter after correction by the E_f -*ANN* model. As shown in Table 6, for the *Buoy* 51000, the E_f -*ANN* model improves the accuracy of wave spectrum calculations by 25.6 %. Among the spectral shape parameters, the greatest improvement is observed for F_p, with an accuracy enhancement of 39.3 %. H_s follows, and E_{max} shows the smallest improvement.

The temporal variations of H_s, F_p, and E_{max} are shown in Fig. 13. Additionally, the specific observed and computed results for spectral parameters at times (a) and (b) in Fig. 8 are shown in Table 7. Taking H_s as an example, the correlation between the calculated wave height and buoy observations before and after model correction is shown in Fig. 14, where panels (a) and (b) represent the results before and after correction, respectively. After correction using the E_f-*ANN* model, the correlation between the computed H_s and the observations significantly improves, with a closer alignment to the perfect computation ($y = x$), and the underestimation is notably reduced. From the above comparisons, it can be concluded that the E_f -*ANN* model effectively corrects the wave spectrum and spectral parameters, significantly improving the simulation accuracy of the wave model.

Furthermore, this method is compared horizontally with existing studies on the accuracy of wave spectrum computation. Zhang et al. (2024) developed a wave spectrum correction approach using a Vision Transformer deep learning model to adjust the COFAST-SWIM wave spectrum. After correction, the correlation between the modeled spectrum and buoy-observed spectrum is 0.833, while the RMSE for significant wave height computation is 0.23 m.

Hwang et al. (2024) proposed a wave spectrum computation model based on the DeCNN method, utilising wave parameters such as wave height and wave period as model inputs. The resulting Pearson correlation coefficients of the computed spectra at different buoy locations range from 0.89 to 0.95, and the RMSE of wave height prediction is 0.21–0.34 m. Wang et al. (2024) correct the WW3-simulated wave

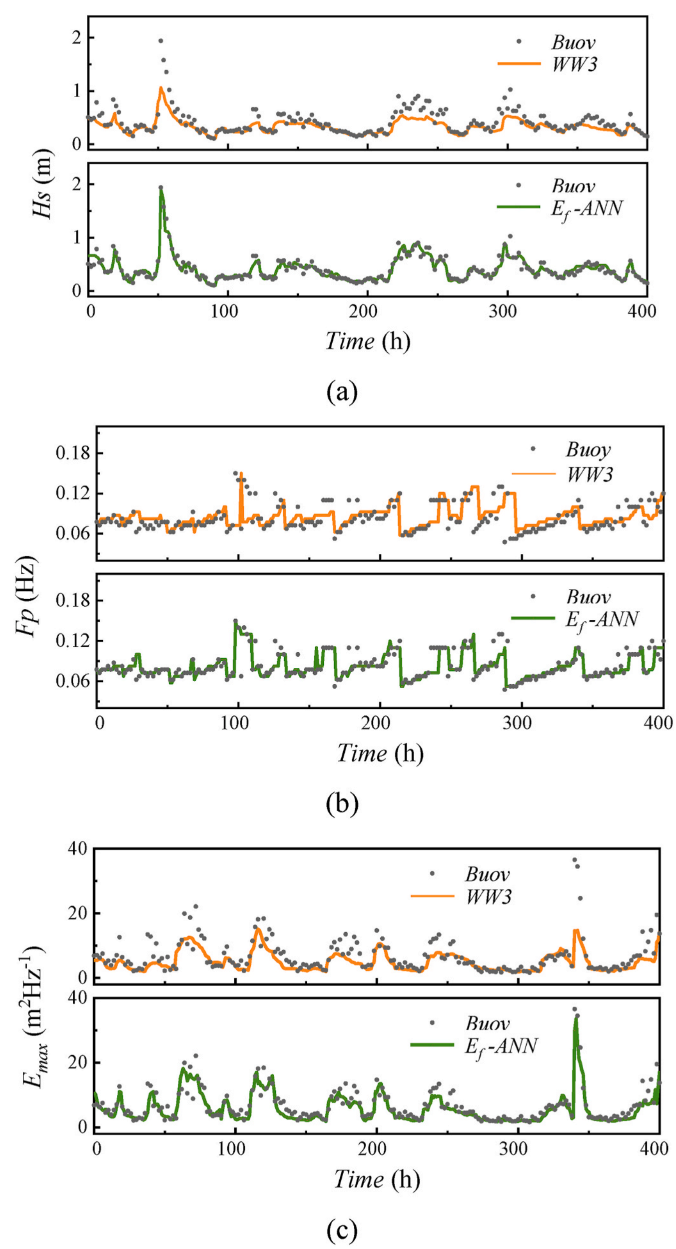

Fig. 13. Comparison of spectral parameter results before and after correction at *Buoy* 51000 with buoy data.

Table 7
Comparison of spectral parameter results before and after correction at *Buoy* 51000.

	Time *a*			Time *b*		
	Buoy	WW3	E_f -ANN	Buoy	WW3	E_f -ANN
H_s (m)	0.489	0.387	0.491	0.151	0.144	0.165
F_p (Hz)	0.078	0.083	0.078	0.130	0.130	0.130
E_{max} (m^2Hz^{-1})	13.050	8.171	12.353	2.370	2.021	2.226

results using wave spectrum data observed by the CFOSAT satellite. After assimilation, the RMSE of the significant wave height is 0.322 m.

In this study, after correction using the E_f *-ANN* model, the Pearson correlation coefficient of the wave spectrum is 0.94, while that of wave height is 0.97. These results indicate that, compared with existing approaches, the E_f *-ANN* model achieves higher computational accuracy.

In addition, compared to traditional data assimilation methods, the E_f *-ANN* model for wave spectrum correction is not constrained by the observation time. Traditional data assimilation methods rely on high-quality real-time wave data, with the assimilation results directly corresponding to the observation data in the time dimension. In contrast, once trained, the E_f *-ANN* model developed in this study can independently correct the wave spectrum. By learning the discrepancies between historical numerical simulation results and observational data, the model compensates for simulation biases caused by simplifications of physical processes or parameterisation errors, thereby improving the accuracy of wave spectrum correction. This model significantly reduces the dependence on real-time observational resources, allowing it to maintain high computational accuracy in wave spectrum correction even when data acquisition is delayed or restricted.

However, the E_f *-ANN* model still has certain limitations. As shown in Fig. 8(d), although the model improves the wave spectrum accuracy, but there are still some differences between the optimised spectrum and actual values at some time due to the input data limitations. Therefore, the model's accuracy depends on the wave model's calculation accuracy. Also, the improvement in E_{max} is slightly smaller than that of the wave spectrum and other parameters.

5.2. Model adaptability

Furthermore, the adaptability of the E_f *-ANN* model to different sea areas is analysed. The model is tested using four buoys (51001#-51004#) within the sea area. Two random time instances are selected for these buoy locations to compare the wave spectrum results before and after model correction. The specific results are shown in Figs. 15–18.

The comparison shows that the E_f *-ANN* model performs well at different locations and effectively corrects the WW3 wave spectrum. As shown in Table 8 and Fig. 19, the E_f *-ANN* model significantly improves the wave spectrum accuracy compared to WW3. The MAPE decreases from 38.5 %, 37.4 %, 41.2 % and 36.9 %–33.3 %, 27.6 %, 29.4 % and 26.5 % at four buoy locations. After correction, the spectral correlation exceeds 0.93 at all locations.

Furthermore, a multi-metric evaluation of the model correction is conducted using spectral shape parameters H_s, F_p, and E_{max}. As shown in Table 9, after correction by the E_f *-ANN* model, the MAPE of H_s does not exceed 12.9 %, the MAPE of F_p does not exceed 10.5 %, and the MAPE of

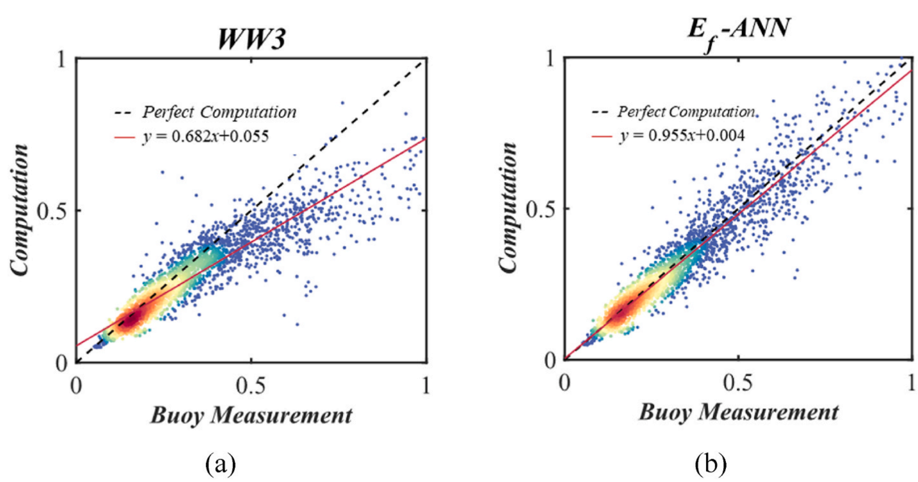

Fig. 14. Correlation comparison of H_s before and after correction at *Buoy* 51000.

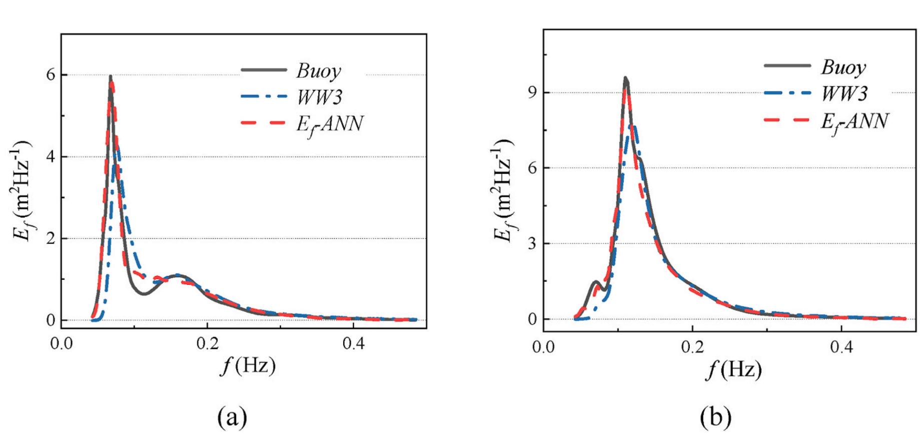

Fig. 15. Wave spectrum comparison before and after correction at *Buoy* 51001.

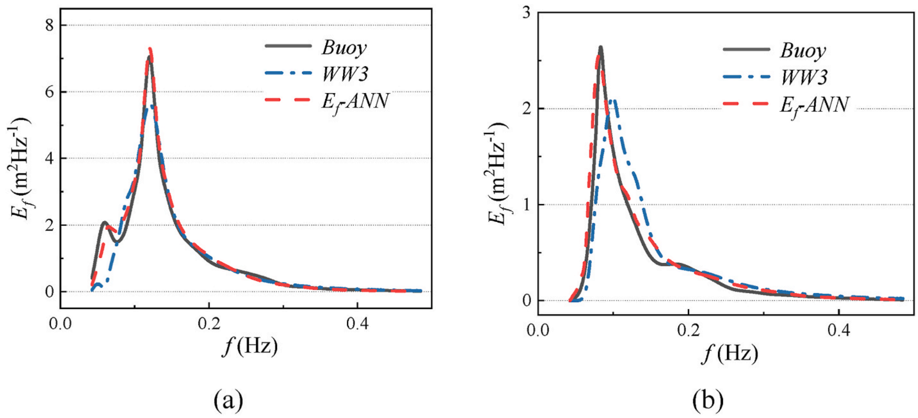

Fig. 16. Wave spectrum comparison before and after correction at *Buoy* 51002.

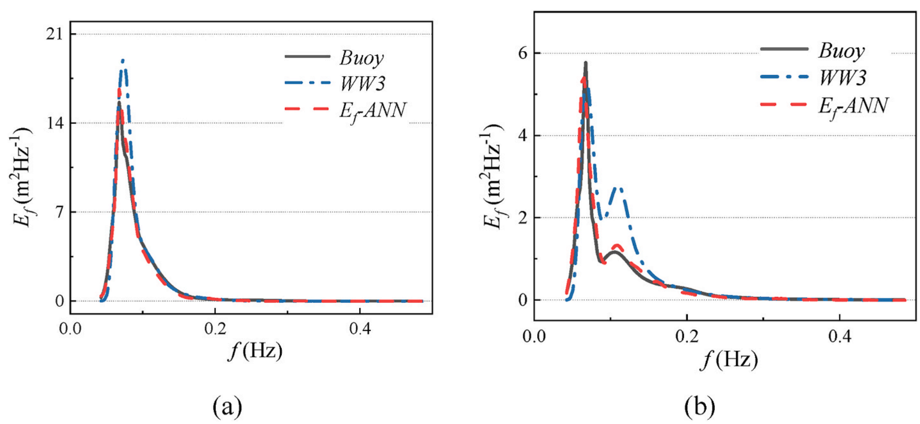

Fig. 17. Wave spectrum comparison before and after correction at *Buoy* 51003.

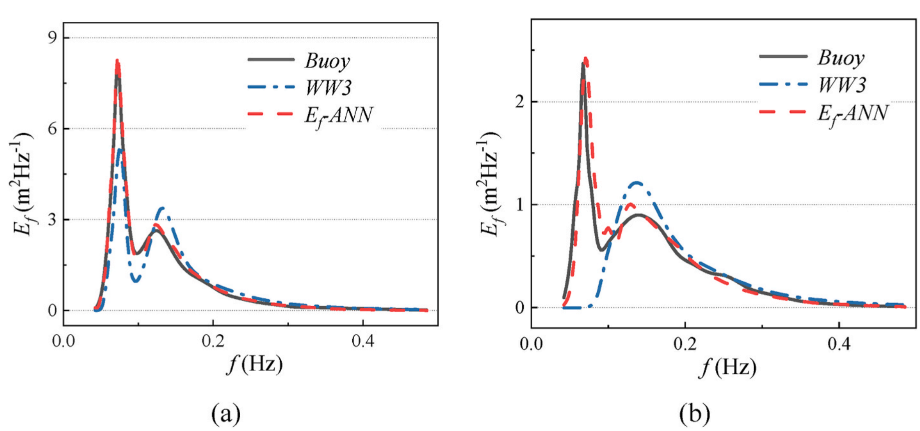

Fig. 18. Wave spectrum comparison before and after correction at *Buoy* 51004.

Table 8
Model adaptability analysis - wave spectrum.

	WW3				E_f -ANN			
	MAE	RMSE	MAPE	r	MAE	RMSE	MAPE	r
51001#	0.75	1.39	38.5 %	0.83	0.48	0.91	33.3 %	0.93
51002#	0.44	0.84	37.4 %	0.80	0.27	0.53	27.6 %	0.94
51003#	0.51	0.92	41.2 %	0.80	0.32	0.61	29.4 %	0.94
51004#	0.48	0.88	36.9 %	0.81	0.28	0.55	26.5 %	0.94

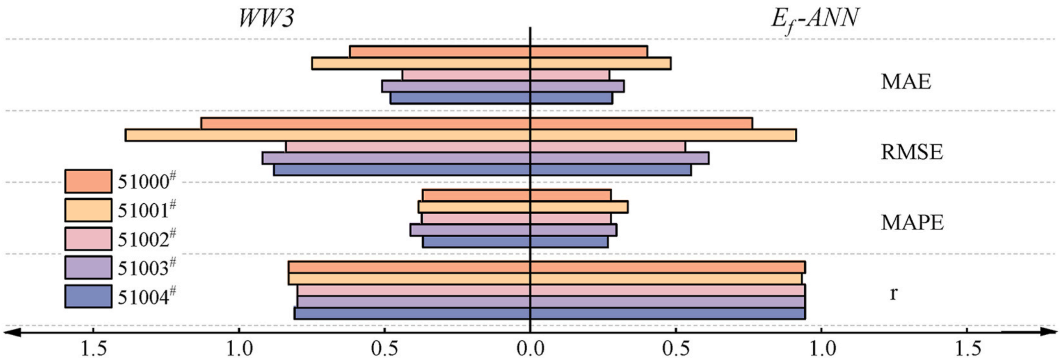

Fig. 19. Model accuracy comparison at multiple locations.

Table 9
Model adaptability analysis - spectral shape parameters.

	H_s		F_p		E_{max}	
	WW3	E_f -ANN	WW3	E_f -ANN	WW3	E_f -ANN
51001#	17.9 %	12.9 %	16.1 %	10.0 %	30.7 %	25.0 %
51002#	14.6 %	10.2 %	20.3 %	10.5 %	29.8 %	24.3 %
51003#	17.9 %	11.4 %	18.0 %	9.9 %	29.7 %	23.5 %
51004#	16.3 %	9.9 %	19.4 %	10.4 %	30.9 %	22.7 %

E_{max} does not exceed 25.0 %. The variation of spectral shape parameters over time is shown in Fig. 20.

Additionally, as shown in Fig. 21 and Table 10, the MAPE of the spectrum E_f at *Buoy* 51001-51004 decreases by more than 25 % overall after model correction. The reduction at location 51001# is relatively low, at 13.4 %. The MAPEs of the spectral parameters H_s, F_p and E_{max} decrease by at least 27.9 %, 37.7 % and 18.7 %, respectively. It can be seen that the model can accurately correct the wave spectrum at different locations within the study area without relying on real-time observational data, thus overcoming the dependency on real-time observations inherent in traditional data assimilation methods. This further enhances the model's adaptability and practicality in oceanographic forecasting applications. The results demonstrate that the E_f *-ANN* model shows good adaptability and can effectively correct the wave spectra at various locations within the sea area.

6. Conclusions

In this study, the E_f *-ANN* model is developed. The model employs deep learning methods to correct the wave spectrum by fitting the discrepancies between numerical simulation results and buoy measurement data. Through the analysis of the model's accuracy and adaptability, the following conclusions are drawn:

(1) The E_f *-ANN* model effectively improves the accuracy of wave spectrum computation and better reproduces the distribution of wave energy. After correction by the E_f *-ANN* model, the MAPE of the wave spectrum decreases from 37.0 % for WW3 to 27.6 %. The spectral correlation increases from 0.83 to 0.94. The percentage decreases in MAPE (δ) for E_f, H_s, F_p, and E_{max} are 25.6 %, 32.3 %, 39.3 % and 14.4 %, respectively.

(2) The E_f *-ANN* model demonstrates good sea area adaptability. Multiple buoy locations within the sea area are selected to test the E_f *-ANN* model. After correction by the E_f *-ANN* model, the MAPE of the wave spectrum decreases from a maximum of 41.2 %–29.4 %, and the correlation coefficient increases from 0.83 to 0.93. For key spectral parameters, the final MAPE for H_s, F_p, and E_{max} are no higher than 12.9 %, 10.5 % and 25.0 %, respectively.

The results demonstrated that the proposed E_f *-ANN* provides a novel

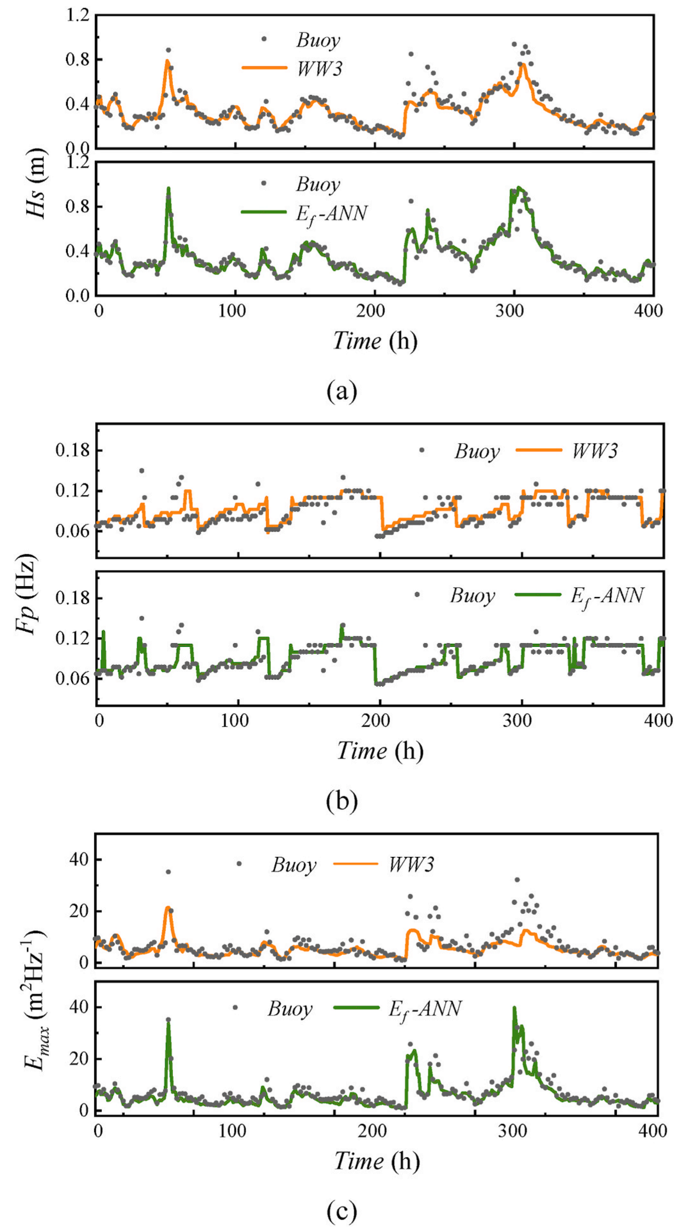

Fig. 20. Comparison of spectral parameter results before and after correction with buoy data at *Buoy* 51001.

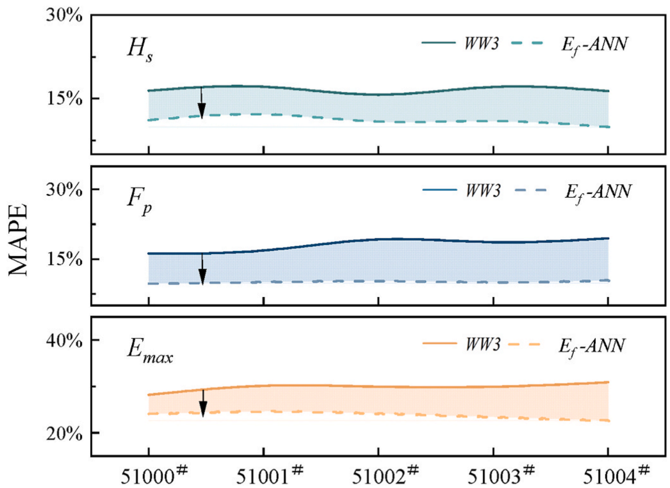

Fig. 21. Improvement in the accuracy of spectral shape parameters.

Table 10
Percentage decrease in MAPE (δ) at different locations.

	E_f	H_s	F_p	E_{max}
51001#	13.4 %	27.9 %	37.7 %	18.7 %
51002#	26.2 %	29.98 %	48.3 %	18.7 %
51003#	28.8 %	36.2 %	45.2 %	21.0 %
51004#	28.1 %	39.2 %	46.4 %	26.5 %

and effective method for wave spectrum computation, and it provides a comprehensive representation of the wave energy distribution.

CRediT authorship contribution statement

Lu Zhang: Writing – original draft, Validation, Methodology, Data curation. **Wenyang Duan:** Writing – review & editing, Methodology. **C. Guedes Soares:** Writing – review & editing. **Jie Zhang:** Writing – review & editing. **Yuliang Liu:** Validation, Resources. **Limin Huang:** Writing – review & editing, Funding acquisition.

Data availability statement

Publicly available datasets were analysed in this study. Data from the ERA5 at ERA5 hourly data on single levels from 1940 to the present, the ETOPO1 at https://ngdc.noaa.gov/mgg/global/relief/ETOPO1/tiled/and the NDBC at https://www.ndbc.noaa.gov/were used in the creation of this manuscript.

Declaration of competing interest

No conflict of interest exits in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

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References

Abbasi, M.R., Sheikh Bahai, A., 2025. The prediction of wind-induced waves in hormuz strait using the SWAN numerical model. Int. J. Coastal, Offshore and Environment. Eng. 10 (1), 35–39.
Ahn, S., Neary, V.S., Allahdadi, M.N., He, R., 2022. A framework for feasibility-level validation of high-resolution wave hindcast models. Ocean Eng. 263, 112193.
Almeida, S., Rusu, L., Guedes Soares, C., 2016. Data assimilation with the ensemble kalman filter in a high-resolution wave forecasting model for coastal areas. J. Operational Oceanograph. 9 (2), 103–114.
Amarouche, K., Akpınar, A., Rybalko, A., Myslenkov, S., 2023. Assessment of SWAN and WAVEWATCH-III models regarding the directional wave spectra estimates based on eastern black sea measurements. Ocean Eng. 272, 113944.
Ardhuin, F., O'Reilly, W.C., Herbers, T., Jessen, P.F., 2003. Swell transformation across the continental shelf. Part I: Attenuation and directional broadening. J. Phys. Oceanogr. 33 (9), 1921–1939.
Bennett, V.C., Mulligan, R.P., 2017. Evaluation of surface wind fields for prediction of directional ocean wave spectra during hurricane sandy. Coast. Eng. 125, 1–15.
Bernardino, M., Rusu, L., Guedes Soares, C., 2021. Evaluation of extreme storm waves in the black sea. J. Operational Oceanograph. 14 (2), 114–128.
Beyramzadeh, M., Siadatmousavi, S.M., 2022. Skill assessment of different quadruplet wave-wave interaction formulations in the WAVEWATCH-III model with application to the Gulf of Mexico. Appl. Ocean Res. 127, 103316.
Bretherton, F.P., Garrett, C.J.R., 1968. Wavetrains in inhomogeneous moving media. Proceed. Roy. Soc. London. Series A. Math. Phys. Sci. 302 (1471), 529–554.
Campos, R.M., D'Agostini, A., Franca, B.R.L., Damiao, A.L.A., Guedes Soares, C., 2022. Implementation of a multi-grid operational wave forecast in the south Atlantic Ocean. Ocean Eng. 243, 110173.
Costa, M.O., Campos, R.M., Guedes Soares, C., 2023. Enhancing the accuracy of Metocean hindcasts with machine learning models. Ocean Eng. 287, 115724.
da Silva, M.B.L., Barreto, F.T.C., de Oliveira Costa, M.C., da Silva Junior, C.L., de Camargo, R., 2025. Bias correction of significant wave height with LSTM neural networks. Ocean Eng. 318, 120015.
Deshmukh, A.N., Deo, M.C., Bhaskaran, P.K., Nair, T.B., Sandhya, K.G., 2016. Neural-network-based data assimilation to improve numerical ocean wave forecast. IEEE J. Ocean. Eng. 41 (4), 944–953.
Elshinnawy, A.I., Antolínez, J.A., 2023. A changing wave climate in the Mediterranean Sea during 58-years using UERRA-MESCAN-SURFEX high-resolution wind fields. Ocean Eng. 271, 113689.
Elshinnawy, A.I., Lobeto, H., Menendez, M., 2024a. Changing wind-generated waves in the Red Sea during 64 years. Ocean Eng. 297, 116994.
Elshinnawy, A.I., Menéndez, M., Medina, R., 2024b. A parameterization for the correction of ERA5 severe winds for extreme ocean wave modelling. Ocean Eng. 312, 119048.
Emmanouil, G., Galanis, G., Kallos, G., 2010. A new methodology for using buoy measurements in sea wave data assimilation. Ocean Dyn. 60 (5), 1205–1218.
Feng, X., Ma, G., Su, S.F., Huang, C., Boswell, M.K., Xue, P., 2020. A multilayer perceptron approach for accelerated wave forecasting in Lake Michigan. Ocean Eng. 211, 107526.
Gemmrich, J., Thomas, B., Bouchard, R., 2011. Observational changes and trends in northeast Pacific wave records. Geophys. Res. Lett. 38 (22).
Hasselmann, S., Hasselmann, K., Allender, J., Barnett, T.P., 1985. Computations and parameterizations of the nonlinear energy transfer in a gravity-wave spectrum. Part II: parameterizations of the nonlinear energy transfer for application in wave models. J. Phys. Oceanogr. 15, 1378–1391.
Hasselmann, S., Lionello, P., Hasselmann, K., 1997. An optimal interpolation scheme for the assimilation of spectral wave data. J. Geophys. Res.: Oceans 102 (C7), 15823–15836.
Henriques, M.R., Silva, D., Yanchin, I., Latas, M., Guedes Soares, C., 2025. Improving the forecast of wind speed and significant wave height using neural networks and gradient boosting trees. Ocean Eng. 327, 120925.
Hwang, S., Lee, J.L., Chun, H., 2024. Applying artificial neural network to zero-crossing wave parameters for the wave spectrum. Ocean Eng. 309, 118331.
Ito, K., Xiong, K., 2000. Gaussian filters for non-linear filtering problems. IEEE Trans. Automat. Control 45 (5), 910–927.
Jamous, M., Marsooli, R., 2023. A multidecadal assessment of mean and extreme wave climate observed at buoys off the US east, gulf, and west coasts. J. Mar. Sci. Eng. 11 (5), 916.
Jiang, D., Huang, B., Miao, Q., Sun, H., Wang, Z., 2025. Extreme design wave parameters optimization of typhoon wave for ocean engineering based on numerical simulation and observation data in the South China Sea. Ocean Eng. 323, 120603.
Kingma, D.P., 2014. Adam: a method for stochastic optimisation. arXiv preprint arXiv: 1412.6980.
Lucas, C., Boukhanovsky, A., Guedes Soares, C., 2011. Modelling the climatic variability of directional wave spectra. Ocean Eng. 38 (11–12), 1283–1290.
Lucas, C., Silva, D., Guedes Soares, C., 2023. Climatic directional wave spectra in coastal sites. Coast. Eng. 180, 104255.
Lucero, F., Stringari, C.E., Filipot, J.F., 2023. Improving WAVEWATCH III hindcasts with machine learning. Coast. Eng. 185, 104381.
McCulloch, W.S., Pitts, W., 1943. A logical calculus of the ideas immanent in nervous activity. Bull. Math. Biophys. 5, 115–133.
Oladejo, H.O., Bernstein, D.N., Cambazoglu, M.K., Nechaev, D., Abdolali, A., Wiggert, J. D., 2025. Wind forcing, source term and grid optimisation for hurricane wave modelling in the Gulf of Mexico. Coast. Eng. 197, 104692.
Ponce de Leon, S., Guedes Soares, C., 2008. Sensitivity of wave model predictions to wind fields in the Western Mediterranean sea. Coast. Eng. 55 (11), 920–929.

Ponce de León, S., Osborne, A.R., 2020. Role of non-linear four-wave interactions source term on the spectral shape. J. Mar. Sci. Eng. 8 (4), 251.
Rusu, L., Guedes Soares, C., 2014. Local data assimilation scheme for wave predictions close to the Portuguese ports. J. Operational Oceanograph. 7 (2), 45–57.
Shankar, C.G., Cambazoglu, M.K., Bernstein, D.N., Hesser, T.J., Wiggert, J.D., 2025. Sensitivity and impact of atmospheric forcings on hurricane wind wave modeling in the Gulf of Mexico using nested WAVEWATCH III. Appl. Ocean Res. 154, 104320.
Stopa, J.E., Ardhuin, F., Babanin, A., Zieger, S., 2016. Comparison and validation of physical wave parameterizations in spectral wave models. Ocean Model. 103, 2–17.
Sun, H., Zhao, W., Yang, Q., Cai, S., Liang, X., Tian, J., 2019. Estimating four-dimensional internal wave spectrum in the northern South China Sea. J. Atmos. Ocean. Technol. 36 (7), 1199–1216.
The WAMDI Group, 1988. The WAM model—A third generation ocean wave prediction model. J. Phys. Oceanogr. 18 (12), 1775–1810.
Tolman, H.L., 1992. Effects of numerics on the physics in a third-generation wind-wave model. J. phys. Oceanogr. 22 (10), 1095–1111.
Van Vledder, G.P., Akpınar, A., 2015. Wave model predictions in the black sea: sensitivity to wind fields. Appl. Ocean Res. 53, 161–178.
Wang, C., Li, S., Yu, H., Wu, K., Lang, S., Xu, Y., 2024. Comparison of wave spectrum assimilation and significant wave height assimilation based on Chinese-French oceanography satellite observation. Remote Sens. Environ. 305, 114085.
Wang, et al., 2024. Corrected the WW3-simulated wave results using wave spectrum data observed by the surface wave investigation and monitoring instrument aboard the China-France Oceanic satellite (CFOSAT) satellite. After Assimilation, the RMSE of Significant Wave Height was 0.322 M.
Whitham, G.B., 1965. Non-linear dispersive waves. Proceed. Roy. Soc. London. Series A. Math. Phys. Sci. 283 (1393), 238–261.
Zhang, L., Duan, W., Wu, K., Cui, X., Guedes Soares, C., Huang, L., 2024. Optimized WAVEWATCH III for significant wave height computation using machine learning. Ocean Eng. 312, 119004.
Zhang, R., Qi, J., Yan, Q., Fan, C., Miao, Q., Yang, Y., Zhang, J., 2024. Improvements to the CFOSAT SWIM wave spectrum based on the ViT deep learning model. IEEE Geosci. Remote Sens. Lett. 21, 1–5.
Zieger, S., Babanin, A.V., Rogers, W.E., Young, I.R., 2015. Observation-based source terms in the third-generation wave model WAVEWATCH. Ocean Model. 96, 2–25.
Zou, X., 2025. Overview and new opportunities for multi-source data assimilation. J. Meteorological Res. 39 (1), 1–25.